



BARKER COLLEGE

**TRIAL HIGHER SCHOOL CERTIFICATE
2000**

**MATHEMATICS
3 UNIT (ADDITIONAL)
AND
3/4 UNIT (COMMON)**

BTP
AES
CFR
PJR
MRB
JGD*
JFH*

PM TUESDAY 1 AUGUST
LORENZO MONTABILLI
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MONTABILLI 2061



*TIME ALLOWED : TWO HOURS
[Plus 5 minutes reading time]*

100 copies

DIRECTIONS TO STUDENTS:

- Write your Barker Student Number on **EACH AND EVERY** page.
- Students are to attempt **ALL** questions.
ALL questions are of equal value. [12 marks]
- The questions are not necessarily arranged in order of difficulty.
Students are advised to read the whole paper carefully at the start of the examination.
- **ALL** necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Begin your answer to each question on a **NEW** page. The answers to the questions in this paper are to be returned in **SEVEN SEPARATE BUNDLES**.
Write on **ONLY ONE SIDE** of each page.
- Approved calculators and geometrical instruments may be used.
- A table of Standard Integrals is provided at the end of the paper.

* * * *

QUESTION 1.(a) Solve for x :

(i) $\frac{x+4}{x-2} > 5$ [3m]

(ii) $\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right) + 6 = 0$ [3m]

(b) Differentiate with respect to x :

(i) $\cos^3 2x$ [2m]

(ii) $e^{x \ln x}$ [2m]

(c) AB is a variable interval. M and N divide AB in ratio $-2 : 1$ and $2 : 1$ respectively.Draw a diagram and decide in what ratio B divides MN . [2]**QUESTION 2.**

(a) Evaluate: $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$ [2m]

(b) (i) Sketch the curve $y = \sin^{-1}(2x)$

(ii) State the domain and range of this function. [3m]

(c) Evaluate: $\int_0^2 \frac{4}{\sqrt{4-x^2}} dx$ [3m]

(d) Find the obtuse angle, to the nearest minute, between the lines

$$3x - 4y + 8 = 0 \text{ and } x + 2y + 1 = 0$$
 [4m]

QUESTION 3.

- (a) Prove: $\frac{\sin \theta + \sin 2\theta}{1 + \cos \theta + \cos 2\theta} = \tan \theta$ [3m]
- (b) By using the substitution $u = \cos x$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \tan x \, dx$ [4m]
- (c) If ${}^9C_4 + {}^9C_5 = {}^{10}C_m$, find the value of m . [1m]
- (d) Find the derivatives of:
- (i) $\ln(\sec 3x)$
- (ii) $\tan^{-1}(2 \tan x)$ [4m]

QUESTION 4.

- (a) $P(4p, 2p^2)$ is a point on the parabola $x^2 = 8y$ and S is the focus. The tangent to the parabola at P meets the y -axis in M . The perpendicular from the focus S to the tangent PM meets the tangent in N .
- (i) Write down the equation of PM and **hence** show that M has coordinates $(0, -2p^2)$. [1m]
- (ii) Write down the equation of SN and **hence** find the coordinates of N . [4m]
- (iii) Find the coordinates of the midpoint of the interval MN . [1m]
- (iv) Find the equation of the locus of the midpoint MN as P varies. [1m]
- (b) Use the binomial theorem to find the term in x^5 in the expansion $(1 + 2x)^8$. [2m]
- (c) Give the exact value of $\cos^{-1}\left(\sin \frac{4\pi}{3}\right)$. [3m]

QUESTION 5.

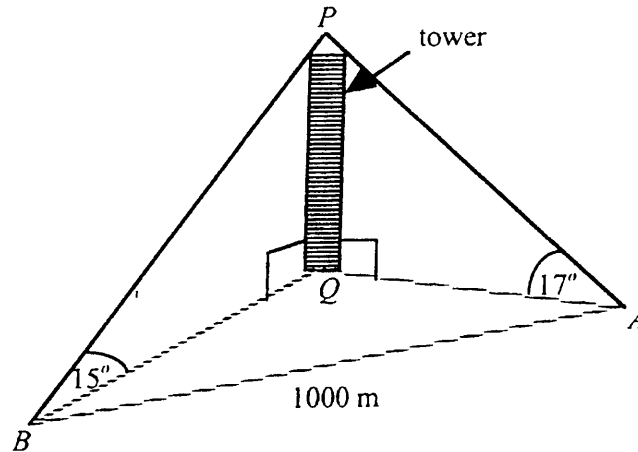
- (a) Prove, by mathematical induction, that $3^{2^n} - 1$ is divisible by 8 for all positive integers. [3m]
- (b) Rain is falling steadily and is collected in an inverted right cone so that the volume collected increases at a constant rate of $5 \text{ cm}^3/\text{h}$. If the radius $r \text{ cm}$ of the surface of the water is one third its depth, $y \text{ cm}$, find the rate in cm/h at which the depth is increasing when $y = 3.5$. [5m]
- (c) Find all angles θ with $0 \leq \theta \leq 2\pi$ for which $\cos 2\theta = \cos \theta$. [4m]

QUESTION 6.

- (a) Find the term independent of x in the expansion of $\frac{1}{x} \left(3x - \frac{1}{2x} \right)^7$. [3m]
- (b) A particle moves in a straight line and its position at any time t is given by:
- $$x = 2 \cos 3t - 5 \sin 3t.$$
- (i) Find the acceleration in terms of position and **hence** show that the motion is simple harmonic.
- (ii) Find the greatest speed of the particle. [5m]
- (c) (i) Show that $\frac{d}{dx} [e^x (\sin x + \cos x)] = 2e^x \cos x$. [4m]
- (ii) **Hence**, evaluate: $\int_1^{\frac{\pi}{2}} e^x \cos x \, dx$ (correct to 3 significant figures). [4m]

QUESTION 7.

(a)



The angle of elevation of a tower PQ , of height h metres, at a point A due east of it, is 17° . From another point B , the bearing of the tower is 061°T and the angle of elevation is 15° . The points A and B are 1000 metres apart and on the same level as the base Q of the tower.

- (i) Show that $\angle AQB = 151^\circ$.
- (ii) Consider the $\triangle APQ$ and show that $AQ = h \tan 73^\circ$.
- (iii) Find a similar expression for BQ .
- (iv) Calculate h , using the cosine rule, in the $\triangle AQB$.
(Answer to nearest metre).

[6m]

- (b) A cricket ball is projected from the ground with an initial velocity of 30 ms^{-1} at an angle of 40° to the horizontal. The equations of motion taken in the horizontal and vertical directions are $\ddot{x} = 0$, $\ddot{y} = -10$. (Use $g = 10 \text{ ms}^{-2}$).

- (i) Calculate the greatest height reached by the ball.
- (ii) What is the speed of the ball at the greatest height?
- (iii) How high is it after the ball has travelled 40 metres horizontally?

[6m]

END OF EXAM

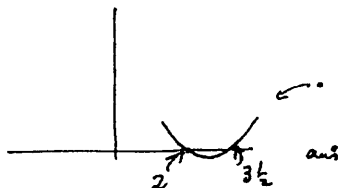
ANS. (3u) Trial 2000.

Q1. (a) (i) Method $1: x(x-2)^2: (x+4)(x-2) > 5(x-2)^2$
 $x^2 + 2x - 8 > 5(x^2 - 4x + 4)$
 $0 > 4x^2 - 22x + 28$

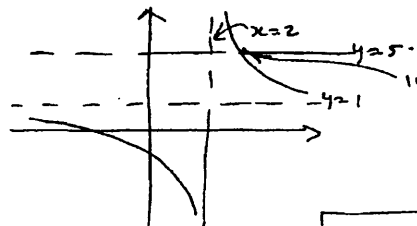
14. $2x^2 - 11x + 14 < 0$

$$(2x - 7)(x - 2) < 0$$

$$\therefore \boxed{2 < x < 3\frac{1}{2}}$$



method 2: sketch $y = \frac{x-2+6}{x-2} = 1 + \frac{6}{x-2}$



$$\begin{aligned} \text{Substituting: } x+4 &= 5(x-2) \\ &= 5x-10 \\ 4x-14 &= 0 \\ x &= \frac{14}{4} = \frac{7}{2} = 3\frac{1}{2} \end{aligned}$$

$$\therefore 2 < x < 3\frac{1}{2}$$

method 3 : cases:

for $x > -2$: $x+4 > 5(x-2) \Rightarrow x+4 > 5x-10$
 $4x-14 < 0 \therefore x < 3\frac{1}{2}$
 $\therefore -2 < x < 3\frac{1}{2}$ is part sol.

for $x < 2$: $x + 4 < 5(x - 2) \dots \Rightarrow x > 3\frac{1}{2}$

no part sol. here

$$\therefore \boxed{2 < x < 3\frac{1}{2}}$$

$$(ii) \quad y^2 - 5y + 6 = 0 \Rightarrow (y-2)(y-3) = 0$$

$$\therefore x + \frac{1}{x} = 2, 3$$

$$x^2 - 2x + 1 = 0 \quad \text{or} \quad x^2 - 3x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = \frac{3 \pm \sqrt{9-4}}{2}$$

$$\therefore x = 1, \frac{3 \pm \sqrt{5}}{2}$$

Q1. (b) (i) $y = \cos^3 2x$

$$y' = 3\cos^2 2x \cdot -\sin 2x \cdot 2 \quad \leftarrow 2 \text{ marks, 1 off each mistake.}$$

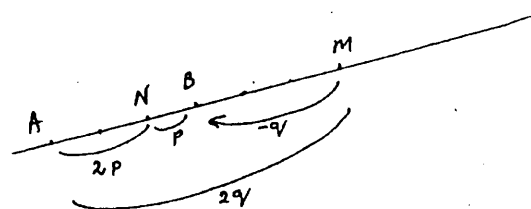
$$= -6\sin 2x \cdot \cos^2 2x$$

(ii) $y = e^{x \ln x}$

$$y' = (1 \cdot \ln x + x \cdot \frac{1}{x}) e^{x \ln x} \leftarrow 2 \text{ marks, 10ft each mistake.}$$

$$= (1 + \ln x) e^{x \ln x}.$$

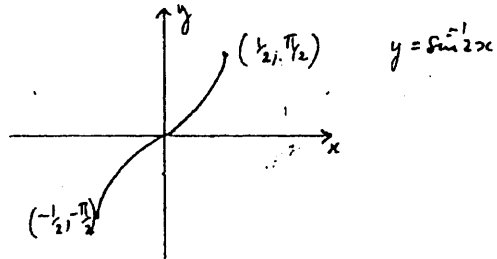
(c)



B divides MN in ratio 3:1

$$Q2. (a) \quad 1 = \lim_{x \rightarrow 0} \frac{\sin 5x}{2x} = \lim_{5x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{5}{2} \\ = 1 \times \frac{5}{2} = 2\frac{1}{2}$$

$$(b) (i) \quad -1 \leq 2x \leq 1 \Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$$



$$(ii) \quad \text{Domain } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \text{Range } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$(c) \quad I = \int_0^2 \frac{4 \, dx}{\sqrt{4-x^2}} = 4 \left[\sin^{-1} \frac{x}{2} \right]_0^2 \\ = 4 \{ \sin^{-1} 1 - \sin^{-1} 0 \} \\ = 4 \{ \frac{\pi}{2} - 0 \} \\ = 2\pi$$

$$(d) \quad m_1 = \frac{3}{4}, \quad m_2 = -\frac{1}{2} \\ \tan \theta = \left| \frac{\frac{3}{4} - (-\frac{1}{2})}{1 + \frac{3}{4}(-\frac{1}{2})} \right| = \frac{5/4}{5/8} = 2 \\ \uparrow \\ \text{acute.} \\ \therefore \theta = 180^\circ - 63^\circ 26' = 116^\circ 34'$$

$$Q3. (a) \quad LHS = \frac{\sin \theta + 2 \sin \theta \cos \theta}{1 + \cos \theta + 2 \cos^2 \theta - 1} \\ = \frac{\sin \theta (1 + 2 \cos \theta)}{\cos \theta} \\ = \tan \theta \\ = RHS.$$

$$(b) \quad u = \cos x, \quad x = \pi/3 \Rightarrow u = \frac{1}{2} \\ du = -\sin x \, dx, \quad x = 0 \Rightarrow u = 1$$

$$I = - \int_0^{\pi/3} \frac{-\sin x \, dx}{\cos x} \\ = - \int_1^{\frac{1}{2}} \frac{du}{u} \\ = \int_{\frac{1}{2}}^1 \frac{du}{u} \\ = [\ln u]_{\frac{1}{2}}^1 \\ = \ln 1 - \ln \frac{1}{2} \\ = 0 - (-\ln 2) \\ = \ln 2$$

$$(c) \quad LHS = \frac{9!}{5!4!} + \frac{9!}{4!5!} = \frac{2 \times 9!}{5!4!} \times \frac{5}{5} = \frac{10!}{5!5!} = {}^{10}C_5 \\ \therefore m = 5 \quad [\text{note bold answer OK}]$$

$$(b) (i) \quad \frac{d}{dx} (\ln(\sec 3x)) = \frac{3 \times \sec 3x \tan 3x}{\sec 3x} \\ = 3 \tan 3x$$

$$(ii) \quad \frac{d}{dx} (\tan^{-1}(2 \tan x)) = \frac{2 \sec^2 x}{1 + 4 \tan^2 x}$$

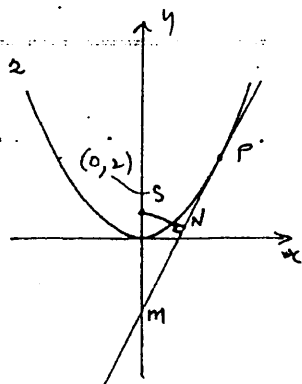
Q4

(a)(i) PM is $x \cdot 4p = 4(y + 2p^2)$ $a = 2$

ie. $px = y + 2p^2$

cuts y axis: $x = 0 \therefore y = -2p^2$

ie. M is $(0, -2p^2)$



(ii) gr. PM = p

$\therefore$ SN is $y - 2 = -\frac{1}{p}(x - 0)$

ie. $y = 2 - \frac{x}{p}$

N: $px = (2 - \frac{x}{p}) + 2p^2$

ie. $p^2x = 2p - x + 2p^3$

$x(p^2 + 1) = 2p(p^2 + 1)$

$\therefore x = 2p$, since $p^2 + 1 > 0$

$\& y = 2 - \frac{2p}{p} = 0$

$\therefore$ N is $(2p, 0)$

(iii) midpt of MN: $(\frac{0+2p}{2}, \frac{-2p^2+0}{2})$

ie. $(p, -p^2)$

(iv) locus: $y = -p^2 = -x^2$

ie. $y = -x^2$

(b) $(1+2x)^8 = \binom{8}{0} + \binom{8}{1}(2x) + \dots + \binom{8}{5}(2x)^5 + \dots + (2x)^8$
 $\therefore$ coeff of x^5 is $\binom{8}{5} \times 2^5 = \frac{8 \times 7 \times 6}{1 \times 2 \times 3} \times 32 = 7 \times 256$
 $= 1792$

(c) $\sin \frac{4\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$, $\cos^{-1}(-\frac{\sqrt{3}}{2})$
 $\therefore$ Ans = $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$

Q5.(a) $n=1 \Rightarrow 3^{2 \times 1} - 1 = 9 - 1 = 8 \therefore$ div by 8 when $n=1$

Say $(n=k)$, $3^{2k} - 1 = 8P$ for some pos. int. k, P

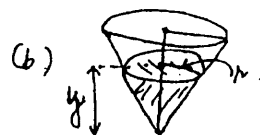
then $3^{2(k+1)} - 1 = 3^{2k+2} - 1$

$= 9 \times (3^{2k} - 1) + 8$

$= 9 \times 8P + 8$

$= 8(9P + 1)$ & $(9P + 1)$ is an int.

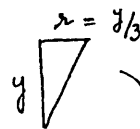
$\therefore$ If div by 8 for some value of n then div by 8 for next value of n and shown true for $n=1 \therefore$ true for all pos. int. n .



$V = \frac{1}{3} \pi r^2 y$

$\frac{dV}{dt} = 5 \text{ cm}^3/\text{s}$

want $\frac{dy}{dt}$ when $y = 3.5$



$\therefore V = \frac{\pi}{3} \cdot \frac{y^3}{3} = \frac{\pi y^3}{9}$

$\frac{dV}{dy} = \frac{\pi y^2}{3}$

And. $\frac{dV}{dt} = \frac{dV}{dy} \cdot \frac{dy}{dt}$

$5 = \frac{\pi y^2}{3} \cdot \frac{dy}{dt}$

$\therefore \frac{dy}{dt} = \frac{5 \times 3}{\pi (3.5)^2}$ at $y = 3.5$

$\approx 1.2 \text{ cm/s}$

(c) $2\cos^2 \theta - \cos \theta - 1 = 0$

$(2\cos \theta + 1)(\cos \theta - 1) = 0$

$\cos \theta = -\frac{1}{2}, 1$

$\theta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 0, 2\pi$

ie. $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi$

$$Q6(a) \quad \frac{1}{x} \cdot \left(\frac{7}{k}\right) (3x)^{7-k} \left(-\frac{1}{2x}\right)^k$$

$$\text{we want } x^{-1} \times x^{7-k} \times x^{-k} = 1 = x^0$$

$$\text{i.e. } 6 - 2k = 0$$

$$\therefore k = 3$$

$$\therefore \left(\frac{7}{k}\right) 3^{7-k} \left(-\frac{1}{2}\right)^k \text{ is the req. term}$$

$$\text{i.e. } -\frac{7 \times 4 \times 5}{1 \times 2 \times 3} \times \frac{3^4}{2^3} = -\frac{35 \times 81}{8}$$

$$\therefore -354 \frac{3}{8}$$

$$(b) \quad x = 2 \cos 3t - 5 \sin 3t$$

$$\dot{x} = -6 \sin 3t - 15 \cos 3t$$

$$\ddot{x} = -18 \cos 3t + 45 \sin 3t$$

$$(i) \therefore \ddot{x} = -900 \text{ which is sttm}$$

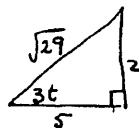
$$(ii) \text{ max speed is when } \ddot{x} = 0$$

$$\text{i.e. } x = 0 \therefore 2 \cos 3t = 5 \sin 3t$$

$$\therefore \frac{2}{5} = \tan 3t$$

$$\& \text{ max speed} = |-6 \times \frac{2}{\sqrt{29}} - 15 \times \frac{5}{\sqrt{29}}|$$

$$= \frac{87}{\sqrt{29}} \doteq 16.155 \doteq 16 \text{ speed units}$$



$$(c) (i) \quad \frac{d}{dx} [e^x (\sin x + \cos x)] = e^x (\sin x + \cos x) + e^x (\cos x - \sin x) = 2e^x \cos x. \quad \text{answ.}$$

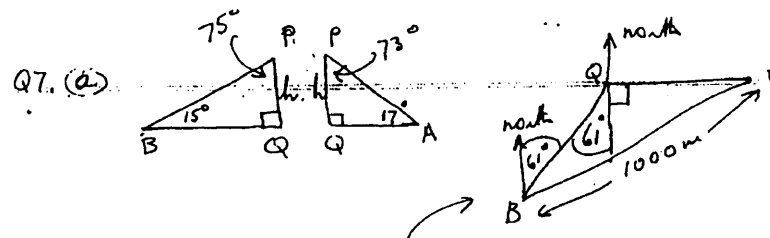
$$(ii) \quad I = \int_1^{\pi/2} e^x \cos x dx$$

$$= \frac{1}{2} \int_1^{\pi/2} 2e^x \cos x dx$$

$$= \frac{1}{2} [e^x (\sin x + \cos x)]_1^{\pi/2}$$

$$= \frac{1}{2} \{e^{\pi/2} (1+0) - e(1+0)\} = \frac{1}{2} \{4.81047 - 2.71828\} = 1.046095$$

$$\doteq 0.527$$



$$(i) \quad \angle AQB = 61^\circ + 90^\circ = 151^\circ$$

$$(ii) \text{ in } \triangle APQ: \tan 73^\circ = \frac{AQ}{h} \therefore AQ = h \tan 73^\circ$$

$$(iii) \text{ in } \triangle BPQ: \tan 75^\circ = \frac{BQ}{h} \therefore BQ = h \tan 75^\circ$$

$$(iv) \text{ in } \triangle ABQ:$$

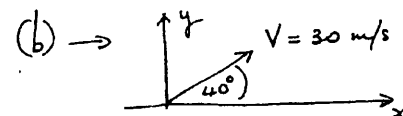
$$1000^2 = (h \tan 73^\circ)^2 + (h \tan 75^\circ)^2 - 2(h \tan 73^\circ)(h \tan 75^\circ) \cos 151^\circ$$

$$= h^2 [\tan^2 73^\circ + \tan^2 75^\circ - 2 \tan 73^\circ \tan 75^\circ \cos 151^\circ]$$

$$= h^2 \times 45.9796 \dots$$

$$\therefore h = \frac{1000}{\sqrt{45.9796 \dots}} = 147.47 \dots$$

$$= 147 \text{ m.}$$



$$\begin{aligned} \ddot{x} &= 0 & \ddot{y} &= -10 \\ \dot{x} &= 30 \cos 40^\circ & \dot{y} &= -10t + 30 \sin 40^\circ \\ x &= 30t \cos 40^\circ & y &= -5t^2 + 30t \sin 40^\circ \end{aligned}$$

$$(i) \text{ max ht: when } \dot{y} = 0: t = 3 \sin 40^\circ$$

$$\text{then ht} = -5(3 \sin 40^\circ)^2 + 90 \sin 40^\circ$$

$$= 45 \sin^2 40^\circ$$

$$\doteq 18.6 \text{ m.}$$

$$(ii) \text{ Speed at top pt} = \dot{x} = 30 \cos 40^\circ \doteq 23 \text{ m/s}$$

$$(iii) \quad x = 40 \Rightarrow t = \frac{40}{3 \cos 40^\circ} \Rightarrow y = -5 \left(\frac{40}{3 \cos 40^\circ} \right)^2 + \frac{40}{3 \cos 40^\circ} \times 30 \sin 40^\circ$$

$$= -16.14743 \dots + 33.56 \dots$$

$$\doteq 17.4 \text{ m.}$$